

Draft: the finite excess in transfinite nim multiplication

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Abstract

Conway's field On_2 makes the ordinals below ω^{ω^ω} an algebraic closure of $\mathbb{F}_2$ under nim-arithmetic. Its multiplication is governed by a Kummer tower whose carries are Lenstra's *excess* values: for each odd prime p , the carry is $\alpha_p = \kappa_{f(p)} + m_p$, where the transfinite part $\kappa_{f(p)}$ has a known closed shape (Lenstra, DiMuro) but the finite correction m_p does not. This note consolidates the state of the excess problem (OPEN.md Problem 3 of `ogdoad`): the exact finite-field reformulation of the root test as a multiplicative-order criterion and its structural norm reduction; a complete analysis of the 3-power column via a half-angle splitting in the cyclotomic component fields, certifying $m_r = 1$ for all primes r with $f(r) \in \{3, 9, \dots, 3^6\}$ and forcing $m_p \geq 4$ unconditionally on the $f(p) = 2 \cdot 3^k$ column; the empirically exact 0/1/4 candidate rule and the proof that the Lenstra component set alone cannot decide the excess; and the reduction of Lenstra's open boundedness question to that candidate. Feasibility of the first open row, $p = 719$, is assessed: a Frobenius-orbit norm recurrence reduces it to arithmetic in $\mathbb{F}_{2^{359}}$, with the cost concentrated in tower-aware Frobenius composition. Each statement carries an explicit claim-level tag.

Claim levels. [**proved**] – unconditional, machine-checked or classically verified; [**certified** $k \leq 6$] – proved for the stated range, blocked beyond it only by unfactored cofactors; [**consistent**] – no counterexample found, not proved; [**conjectured**] – explicitly open in the literature or posed in this thread; [**open**] – neither confirmed nor refuted, no feasible attack on record.

1 Notation, implemented state, external data

1.1 Notation

For an odd prime p :

- $f(p) = \text{ord}_p(2)$, the multiplicative order of 2 modulo p ;
- κ_h is the Lenstra/DiMuro tower element indexed by h ;
- $\mathcal{Q}(h)$ is Lenstra's set of prime-power components appearing in κ_h ;
- the *excess* m_p is the least finite m such that $\kappa_{f(p)} + m$ has no p -th root in the relevant finite component field;
- the Kummer carry is $\alpha_p = \kappa_{f(p)} + m_p$.

For the ordinal tower, a row with $\mathcal{Q}(f(p)) = \{q\}$ and finite excess m gives the ordinal sum corresponding to $\kappa_q + m$. Addition of nimbers is XOR; $\otimes$ is the nim product.

1.2 Implemented tower state

In the Rust tower (`src/scalar/big/ordinal/tower.rs`):

- DiMuro Table 1 rows [3] through α_{43} ;
- a locally verified row $\alpha_{47} = \omega^{(\omega^7)} + 1$ (Section 7);
- the operational boundary is α_{53} : a carry needing α_{53} or beyond returns `None`.

1.3 External data snapshot (2026-06-09)

- OEIS A380496 [7] has 1417 extended rows: 799 known and 618 unknown; the b-file has 126 initial known rows.
- The first OEIS unknown is row $n = 127$, the 127th odd prime $p = 719$. For $p = 719$: $f(719) = \text{ord}_{719}(2) = 359$ and $\mathcal{Q}(359) = \{359\}$.
- The transfinite-nim-calculator logs [9] record the direct component exponent as $e_{719} = 1,258,230,380$, the practical wall for direct exponentiation.

2 The order criterion and structural norm reduction

2.1 The exact criterion

Throughout, E denotes the degree of the component field containing $\beta = \kappa_{f(p)} + m$, so $f = f(p) \mid E$ and hence $p \mid 2^E - 1$.

Proposition 1 (power criterion). *[proved] $\beta \in \mathbb{F}_{2^E}^\times$ has no p -th root in $\mathbb{F}_{2^E}$ iff $\beta^{(2^E-1)/p} \neq 1$. Hence*

$$m_p = \min\{m \geq 0 : (\kappa_{f(p)} + m)^{(2^E-1)/p} \neq 1\}.$$

Proof. $\mathbb{F}_{2^E}^\times$ is cyclic of order $2^E - 1$ and $p \mid 2^E - 1$; in a cyclic group of order N , β is a p -th power iff $\beta^{N/p} = 1$. $\square$

Corollary 1 (order form). *[proved] If $v_p(2^E - 1) = 1$, then β has no p -th root in $\mathbb{F}_{2^E}$ iff $p \mid \text{ord}(\beta)$, so m_p is the least m with $p \mid \text{ord}(\kappa_{f(p)} + m)$.*

Remark 1 (Wieferich caveat). In general “no p -th root” is equivalent to $v_p(\text{ord } \beta) = v_p(2^E - 1)$, which collapses to $p \mid \text{ord}(\beta)$ only when $v_p(2^E - 1) = 1$. Since $v_p(2^E - 1) = v_p(2^{f(p)} - 1)$ whenever $p \nmid E/f(p)$ (lifting the exponent), the hypothesis can fail only at base-2 Wieferich-type primes with $p^2 \mid 2^{f(p)} - 1$; the only known instances are $p = 1093$ and $p = 3511$, both of which lie inside the extended A380496 range. For those rows the exact test of Proposition 1 (or the full- p -part order condition) must be used. The certifications of Section 3 compute orders exactly, with squarefreeness checks, and are unaffected. The research-note predecessor of this document stated only the order form; the power criterion is the correct general statement.

2.2 Norm reduction

Proposition 2 (norm reduction). *[proved] For $\beta \in \mathbb{F}_{2^E}^\times$ with $f = f(p) \mid E$,*

$$\beta^{(2^E-1)/p} = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta)^{(2^f-1)/p}.$$

Proof. $N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta) = \beta^{(2^E-1)/(2^f-1)}$, and $((2^E-1)/(2^f-1)) \cdot ((2^f-1)/p) = (2^E-1)/p$. $\square$

So the $p = 719$ test reduces to: (1) compute $N = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^{359}}}(\kappa_{359} + 1)$ in $\mathbb{F}_{2^{359}}$; (2) test whether $719 \mid \text{ord}(N)$ in $\mathbb{F}_{2^{359}}^\times$, a field of degree 359 in which $719 \mid 2^{359} - 1$.

Proposition 3 (Galois-conjugacy invariance). *[proved] The order of $\kappa_{f(p)} + m$ is constant on its Frobenius orbit in $\mathbb{F}_{2^E}$, and the norm $N = \prod_{i=0}^{E/f-1} \text{Frob}^{if}(\kappa_f + m)$ is determined structurally by the tower equations without materialising the full orbit.*

Proposition 4 (tower pinning). *[proved] The tower equations pin κ_{359} up to Frobenius conjugacy: each composed minimal polynomial along the component chain $359 \rightarrow 179 \rightarrow 89 \rightarrow 11 \rightarrow 5 \rightarrow \text{finite}$ is irreducible (the non- q -th-power property defining the lower excess guarantees this), so κ_{359} is a well-defined conjugacy class and N is a well-defined element of $\mathbb{F}_{2^{359}}$.*

2.3 Feasibility for $p = 719$

The component field degree is $E = 2 \cdot 2 \cdot 5 \cdot 11 \cdot 89 \cdot 179 \cdot 359 = 1,258,230,380$; the norm requires $E/f = 3,504,820$ Frobenius steps.

[consistent] The norm can in principle be computed by the binary-splitting Frobenius recurrence $N_{2k} = N_k \cdot \text{Frob}^{f^k}(N_k)$ (iterated-Frobenius technique, von zur Gathen–Shoup [10], Kaltofen–Shoup [4]), requiring about $\log_2(3,504,820) \approx 22$ squarings-with-Frobenius-composition at degree E over $\mathbb{F}_2$. Each Frobenius composition is a modular composition in $\mathbb{F}_2[x]$ modulo a degree- E polynomial; with dense polynomial arithmetic at $E \approx 1.26 \times 10^9$ this is not locally feasible.

[conjectured] (as a cost model, not a theorem) Kummer-tower-aware Frobenius – standard-lattice arithmetic for compatibly embedded finite fields in the style of De Feo–Randriam–Rousseau [2], where q -power Frobenius acts near-monomially per tower level – is the likely 10–100 $\times$ lever: the sparse factored structure $E = 2 \cdot 2 \cdot 5 \cdot 11 \cdot 89 \cdot 179 \cdot 359$ admits a stepwise Frobenius representation in which each level contributes a small-degree modular composition. A *measured* cost model, not a run, is the correct first deliverable.

[open] Whether the Kummer-tower Frobenius model makes the m_{719} computation feasible in wall-clock hours rather than months on available hardware (a single consumer GPU; GF(2) arithmetic is XOR-sliced and GPU-friendly).

Remark 2 (shared abstraction). The structural primitive needed here – $N_{E/K}(\beta) = \prod_i \text{Frob}^i(\beta)$ computed by a Frobenius-orbit recurrence – is the same primitive that the cyclic-algebra Brauer-invariant bridge (Bridge K, `roadmap/TBD.md`) requires for reduced norms over a Frobenius-generated cyclic extension. The existing `FieldExtension::relative_norm` does not apply to the term-algebra/transfinite setting, but the shape is identical; factoring out a reusable `relative_norm_over_frobenius_orbit` is an engineering opportunity. This is not a claim that the bounded `Fpn` norm certifies m_{719} .

3 The 3^k family

Write $\zeta := \kappa_{3^k}$ and $h := 3^k$ throughout this section.

3.1 Structure

Proposition 5 (cyclotomic recognition). *[proved] $\zeta^{3^k} = \kappa_2$, which has order 3; hence ζ is a primitive 3^{k+1} -st root of unity. Since 2 is a primitive root modulo 3^{k+1} , the cyclotomic polynomial*

$\Phi_{3^{k+1}}(x) = x^{2^h} + x^h + 1$ is irreducible over $\mathbb{F}_2$, the component field is $\mathbb{F}_2(\zeta) = \mathbb{F}_{2^{2h}}$, and it carries the index-2 subfield $L = \mathbb{F}_{2^h}$.

Proposition 6 (half-angle splitting). *[proved] With $s = (3^{k+1} + 1)/2$ (the inverse of 2 modulo 3^{k+1}),*

$$\kappa_{3^k} + 1 = \zeta^s (\zeta^s + \zeta^{-s}),$$

where ζ^s lies in the norm-one circle U (of order exactly 3^{k+1} within the subgroup of order $2^h + 1$) and $(\zeta^s + \zeta^{-s})^2 = \zeta + \zeta^{-1} =: \gamma_k \in L^\times$. Since $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$ with coprime orders,

$$\text{ord}(\kappa_{3^k} + 1) = 3^{k+1} \cdot \text{ord}(\gamma_k), \quad \text{ord}(\gamma_k) \mid 2^{3^k} - 1.$$

Machine-checked corollaries: $(\kappa + 1)^{2^h - 1} = \zeta^{-1}$ and $N_{F/L}(\kappa + 1) = \gamma_k$.

Proposition 7 (translates $m = 2, 3$). *[proved] $\kappa_{3^k} + 2$ and $\kappa_{3^k} + 3$ split as in Proposition 6 with γ_k replaced by a Galois conjugate. Therefore $\text{ord}(\kappa_{3^k} + m) = 3^{k+1} \cdot \text{ord}(\gamma_k)$ for $m = 1, 2, 3$, and $\text{ord}(\kappa_{3^k}) = 3^{k+1}$.*

Proposition 8 (the $2 \cdot 3^k$ exception, unconditionally). *[proved] Any prime p with $f(p) = 2 \cdot 3^k$ satisfies $p \mid 2^{3^k} + 1$, hence p divides neither 3^{k+1} nor $2^{3^k} - 1$, hence p never divides $\text{ord}(\kappa_{3^k} + m)$ for $m \in \{0, 1, 2, 3\}$. By the order criterion, $m_p \geq 4$. The full primitivity conjecture C_k below is not needed; the splitting alone forces the bound. The argument applies beyond the current tables, e.g. $p = 87211$, $f = 54$.*

Proposition 9 (the 3-power column). *[proved] For a prime r with $f(r) = 3^k$, the values $m \in \{0, 2, 3\}$ are impossible: $m_r = 0$ would require $r \mid \text{ord}(\kappa_{3^k}) = 3^{k+1}$, a power of 3, while $r \equiv 1 \pmod{3^k}$ is a prime distinct from 3; and $m = 2, 3$ share $m = 1$'s order class by Proposition 7. Any failure of C_k therefore gives $m_r \geq 4$ directly.*

Proposition 10 (norm tower). *[proved] $\gamma_{k-1} = \gamma_k^3 + \gamma_k = N_{L_k/L_{k-1}}(\gamma_k)$ (minimal polynomial $X^3 + X + \gamma_{k-1}$), so $\text{ord}(\gamma_{k-1}) \mid \text{ord}(\gamma_k)$. With lifting-the-exponent $(v_r(2^{3^k} - 1) = v_r(2^{f(r)} - 1))$ for old primes $r \neq 3$, full-order parts propagate upward, and conjecture C_k reduces to C_{k-1} plus primitivity at the new primes $r \mid \Phi_{3^k}(2)$.*

Conjecture 1 (C_k : γ -primitivity). γ_k is primitive in $L^\times = \mathbb{F}_{2^{3^k}}^\times$.

Proposition 11 (equivalence). *[proved] For primes r with $f(r) \mid 3^k$ a 3-power:*

$$C_k \iff m_r = 1 \text{ for every prime } r \text{ with } f(r) \in \{3, 9, \dots, 3^k\}.$$

The family formula is the 0/1/4 candidate of Section 4 restricted to the 3-power column.

3.2 Certification status

[certified $k \leq 6$] C_k holds for $k = 1, \dots, 6$: γ_k is primitive in $\mathbb{F}_{2^{3^k}}^\times$, and $\text{ord}(\kappa_{3^k} + 1) = 3^{k+1} (2^{3^k} - 1)$ is verified exactly – product reconstruction, squarefreeness, primality by deterministic Miller–Rabin below 3.3×10^{24} and MR-64 probable-primality above, an independent sieve re-deriving the small factors, cross-validation against the independent term algebra for $k = 1, 2$ and against the calculator rows $m_{2593} = 1$, $m_{487} = 1$ (experiments/cyclotomic_3k_family.py, experiments/excess/).

[consistent] $k = 7, 8$: every known prime factor of $\Phi_{3^7}(2)$ and $\Phi_{3^8}(2)$ divides $\text{ord}(\gamma_k)$. Full certification is blocked only by the unfactored cofactors (FactorDB status composite-no-factor); no failure has been found.

[open] Whether ECM or GNFS can factor the $\Phi_{3^7}(2)$ and $\Phi_{3^8}(2)$ cofactors within a realistic budget, converting $k = 7, 8$ from CONSISTENT to CERTIFIED.

3.3 Certified excess rows produced

The following $m_r = 1$ rows are certified by the order criterion independently of the cofactor issue (analysis-level results; they are not new Rust carries):

f	primes r with $m_r = 1$
27	262657
81	71119, 97685839
243	16753783618801, 192971705688577, 3712990163251158343
729	80191, 97687, 379081, P42, P90
2187	39367, 7606246033, 263196614521, 529063556041
6561	209953, 1299079, 70063267397606709277393

(P42, P90 denote the certified/probable primes of 42 and 90 decimal digits dividing $\Phi_{3^6}(2)$.)

4 The 0/1/4 candidate

Conjecture 2 (the 0/1/4 rule). *[consistent] (not proved)*

$$m_p = \begin{cases} 0 & \text{if } \mathcal{Q}(f(p)) \text{ is not a singleton odd prime-power,} \\ 4 & \text{if } f(p) = 2 \cdot 3^k, k \geq 1, \\ 1 & \text{otherwise (singleton odd prime-power } \mathcal{Q}(f(p))). \end{cases}$$

The rule matches all 950 calculator records with known $\mathcal{Q}$ -sets and every OEIS-known row covered by those $\mathcal{Q}$ -sets.

4.1 The component set alone is insufficient

[proved] $\mathcal{Q}(f(p))$ alone does not determine m_p :

$\mathcal{Q}$	$m = 4$	$m = 1$
$\{9\}$	$p = 19, f = 18$	$p = 73, f = 6$
$\{81\}$	$p = 163, f = 162$	$p = 2593, f = 18$
$\{243\}$	$p = 1459, f = 486$	$p = 487, f = 18$

[proved] The split for $\mathcal{Q} = \{9\}$ is explained by the order criterion: $\text{ord}(\kappa_9 + 1) = 3^3 \cdot (2^9 - 1) = 27 \cdot 511 = 13797$ and $511 = 7 \cdot 73$, so $73 \mid \text{ord}(\kappa_9 + 1)$ but $19 \nmid \text{ord}(\kappa_9 + 1)$; adding $m = 4$ changes the order and introduces the factor 19.

[proved] The lower bound of the exception column is unconditional: $m_p \geq 4$ for all p with $f(p) = 2 \cdot 3^k$ (Proposition 8).

[conjectured] The matching upper bound $m_p \leq 4$ on that column; equivalent to C_k for the primes in the exception column.

4.2 The $m = 4$ upper bound: the unexplored gap

[open] Whether $m_p = 4$ exactly (not merely ≥ 4) for all primes with $f(p) = 2 \cdot 3^k$. The $m = 4$ translate $\kappa_{3^k} + 4$ involves the number $4 \in \mathbb{F}_{16}$, which does not lie in $\mathbb{F}_2(\zeta)$ (as $4 \nmid 2 \cdot 3^k$); the translate lives in the degree-4 $\cdot 3^k$ compositum, where

$$N_{\mathbb{F}_{4 \cdot 3^k}/\mathbb{F}_{2 \cdot 3^k}}(\kappa_{3^k} + 4) = (\kappa + 4)(\kappa + \text{Frob}^2(4)) = (\kappa + 4)(\kappa + 6),$$

and the same unit-circle/subfield toolkit applies one level up. A proof that this norm has order divisible by every p with $f(p) = 2 \cdot 3^k$ would make $m \leq 4$ rigorous on all 3-adic columns (modulo γ -primitivity). A failure of the analogous splitting is the precise location where an $m \geq 5$ counterexample mechanism would appear.

5 Boundedness

[**conjectured**] (Lenstra [5]) m_p is globally bounded. Lenstra gave unconditional lower-bound rules (singleton odd $\mathcal{Q}$ forces positive excess; $f(p) = 2 \cdot 3^k$ forces excess at least 4) but left absolute boundedness open.

[**proved**] If the 0/1/4 candidate holds, then $m_p \leq 4$ for all p ; proving Conjecture 2 would settle boundedness.

[**proved**] The 3-power column cannot produce $m \in \{2, 3\}$ (Proposition 9); any C_k failure jumps directly to $m_r \geq 4$.

[**proved**] (conditional on C_k) For every prime r with $f(r) = 3^j$, $j \leq k$: C_k implies $m_r = 1$. With C_k certified for $k \leq 6$, all primes in the 3-power column with $f(r) \leq 6561$ have $m_r = 1$ unconditionally.

[**open**] Boundedness outside the 3-power and $2 \cdot 3^k$ columns. The non-cyclotomic singleton chains (e.g. the 11-chain and the 23, 29, 47 components) have $m = 1$ rows in the calculator data and OEIS, but no order-criterion proof covers them the way C_k covers the 3-power column.

[**open**] Whether any prime p has $m_p \geq 5$. No counterexample is known; the question is not known to be decidable by any current structural argument outside the 3-adic columns.

6 Ranked next moves

1. *Dress rehearsal on m_{89}, m_{179}, m_{359} (highest priority).* Certify by direct order tests the calculator rows on which a m_{719} certificate depends. The component fields have degrees $E = 19,580$ ($p = 89$) and $E = 3,504,820$ ($p = 359$ – a stretch: hours to a day with **gf2x**-style arithmetic). This calibrates the cost model for the next move on the same code path and settles the provenance question (“what evidence is acceptable”) before any new α ships. Confirm $m_{359} = 1$ first; the norm recurrence for $p = 719$ shifts if not. *Feasible now; blocks the next move.*
2. *Certify m_{719} via the Kummer-tower Frobenius norm.* Implement the binary-splitting norm $N_{2k} = N_k \cdot \text{Frob}^{359k}(N_k)$ in the tower-sparse representation; measure the cost model on the calibrated path; if tower-aware Frobenius brings the ~ 44 modular compositions to feasible size, run to certify or falsify $m_{719} = 1$. Factor out the shared **relative_norm_over_frobenius_orbit** abstraction (Remark 2). *Blocked on move 1 for the cost model.*
3. *Prove or falsify $m_p = 4$ exactly on the $2 \cdot 3^k$ column.* Apply the half-angle toolkit at the compositum level to $(\kappa+4)(\kappa+6)$ (Section 4); a splitting failure here is where any $m \geq 5$ example would hide. *Feasible with the existing GF(2) toolkit; extends `experiments/cyclotomic_3k_family.py`.*
4. *Convert C_7, C_8 from CONSISTENT to CERTIFIED.* ECM/GNFS jobs on the cofactors of $\Phi_{37}(2)$ and $\Phi_{38}(2)$; full factorization also emits a batch of new A380496 rows. *Background job; uncertain outcome; blocks nothing.*
5. *Exact-order census on the non-cyclotomic singleton chains.* Compute $\text{ord}(\kappa_q + m)$ for small m in the explicit component fields for the 11-chain and the 23, 29, 47 components (degrees up to

~ 5060 ; trivial locally); look for an analogue of the $L^\times \times U$ splitting and for any forced-translate mechanism. This is the main structural gap for boundedness outside the 3-power columns. *Feasible now; no structural theory yet.*

7 Provenance and validation

7.1 Independent oracle

`experiments/ordinal_excess_probe.py` is a small local term-algebra oracle – not a replacement for CGSuite [8] or the C++ calculator [9], but a way to verify the first subtle cases without using the Rust production tower as its own oracle. It implements the impartial term algebra used by the calculators, a multiplicative-order test for small component fields, and a fixed-base exponentiation path (ported from the calculator’s strategy) for targeted root tests where full order factorization is unnecessary. Current probe output:

p	m	$\mathcal{Q}$	root?
7	0	$\{3\}$	yes
7	1	$\{3\}$	no
19	1	$\{9\}$	yes
19	4	$\{9\}$	no
73	1	$\{9\}$	no
47	1	$\{23\}$	no

The last line certifies $m_{47} = 1$ using only lower verified rows; since $f(47) = 23$ and $\mathcal{Q}(23) = \{23\}$, the carry is $\alpha_{47} = \kappa_{23} + 1 = \omega^{(\omega^7)} + 1$, the value now implemented in `tower.rs`.

7.2 The α_{47} promotion

The shipped state: `src/scalar/big/ordinal/tower.rs` carries α_{47} with the landmark test `locally_verified_alpha_47_landmark` and moves the refusal boundary to α_{53} ; the documented boundaries in `nim.rs`, `mod.rs`, `src/games/nimber_game.rs`, `README.md`, and `OPEN.md` were updated in step. The full gate ran clean: `cargo fmt --check`, `cargo test`, `cargo check/clippy` with and without the `python` feature (warnings denied), and the probe under `py_compile`. Focused tests: `locally_verified_alpha_47_landmark` and `boundary_returns_none_past_prime_47`.

7.3 Supporting probes

The 3^k -family certification chain is `experiments/cyclotomic_3k_family.py` (committed, maintained) with the research-run probes under `experiments/excess/` (machine-generated, rescued from ephemeral storage; triage before citing). The analysis in Sections 2–5 synthesizes the 2026-06-10 parallel research run; each claim above carries its own tag, and no claim depends on an unverified solver.

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